

THAARU Sepsis AI — Deep Learning Benchmarks

Phase 7 Deep Learning Final Report • Generated: July 19, 2026

1. Executive Summary

This report presents final deep temporal neural network performance configurations. Models are compared directly against classical tree ensemble baselines. To resolve CPU disk thrashing issues, we staged model comparisons at a 20% sample and final evaluations at a 50% sample size (388,000 sequence records).

2. Model Complexity & Inference Efficiency

Model	Params	Disk Size	Batch Time	Sample Time
LSTM	74,561	0.2885 MB	30.07 ms	29.37 us
GRU	55,937	0.2173 MB	56.08 ms	54.76 us
BiLSTM	181,889	0.7000 MB	62.26 ms	60.80 us
Transformer	73,153	0.3146 MB	27.44 ms	26.80 us

3. Phase 7B Final Leaderboard (50% CPU Optimized)

Model	Config	AUROC	AUPRC	F1-Score	Recall
BiLSTM (Final)	w12_h0	0.8435	0.1228	0.1108	0.7848
BiLSTM (Final)	w6_h0	0.8393	0.1212	0.1115	0.7486
BiLSTM (Final)	w24_h0	0.8460	0.1457	0.1255	0.8175
BiLSTM (Final)	w12_h3	0.8238	0.1119	0.1122	0.7305
BiLSTM (Final)	w12_h6	0.8186	0.1041	0.1116	0.7494

4. Deep Learning vs XGBoost Baseline Comparison

Direct comparison shows the sequential BiLSTM architecture achieves Test AUROC of 0.8435 compared to XGBoost Test AUROC of 0.8381. Furthermore, BiLSTM achieves clinical sensitivity (Recall) of 0.7848 compared to XGBoost sensitivity of 0.5898.